Part IV. Infrastructure Design

This section explores how SREs build and maintain the foundational environments in which services operate. Rather than treating infrastructure as a static set of resources, SRE applies a software engineering mindset to design environments that are inherently robust, scalable, and efficient.

We begin with Chapter 13, "Automation and Tooling". A core tenet of SRE is that we should engineer our systems rather than just operate them, using automation to save time, ensure consistency, and achieve sublinear scaling. This chapter explores that idea, along with when *not* to automate. It discusses the ironies of automation, warning against the deskilling of operators and the creation of new, more complex maintenance work, and provides a framework for incremental automation and systematic toil reduction to keep engineering workloads sustainable.

Next, Chapter 14, "Testing", examines how we build confidence in our systems. It goes beyond the traditional hierarchy of unit, integration, and system tests, exploring advanced techniques such as fault injection testing (chaos engineering) to proactively introduce failures and observe system resilience. The chapter also details using formal methods to mathematically verify system behaviors and catch complex design flaws before they reach production.

In Chapter 15, "Capacity Planning", we examine the delicate balancing act between predicting user demand and managing resource supply. This chapter explores how to achieve capacity efficiency by provisioning for the peak of sums rather than the sum of peaks across diverse workloads. It also provides strategies for building crumple zones into your infrastructure through capacity isolation and traffic management techniques such as criticality-based load shedding, making sure that your most important traffic survives when demand outstrips supply.

Unlike other system failures, data loss is often permanent and carries immense reputational and legal risks. Chapter 16, "Data Management", introduces the hierarchy of data needs, modeled on Maslow's hierarchy of needs, which prioritizes data integrity, security, availability, and efficiency. It emphasizes that backup strategies must be driven

strictly by business-defined SLOs and reminds us that taking backups is meaningless if you aren't regularly and automatically testing your restores.

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